

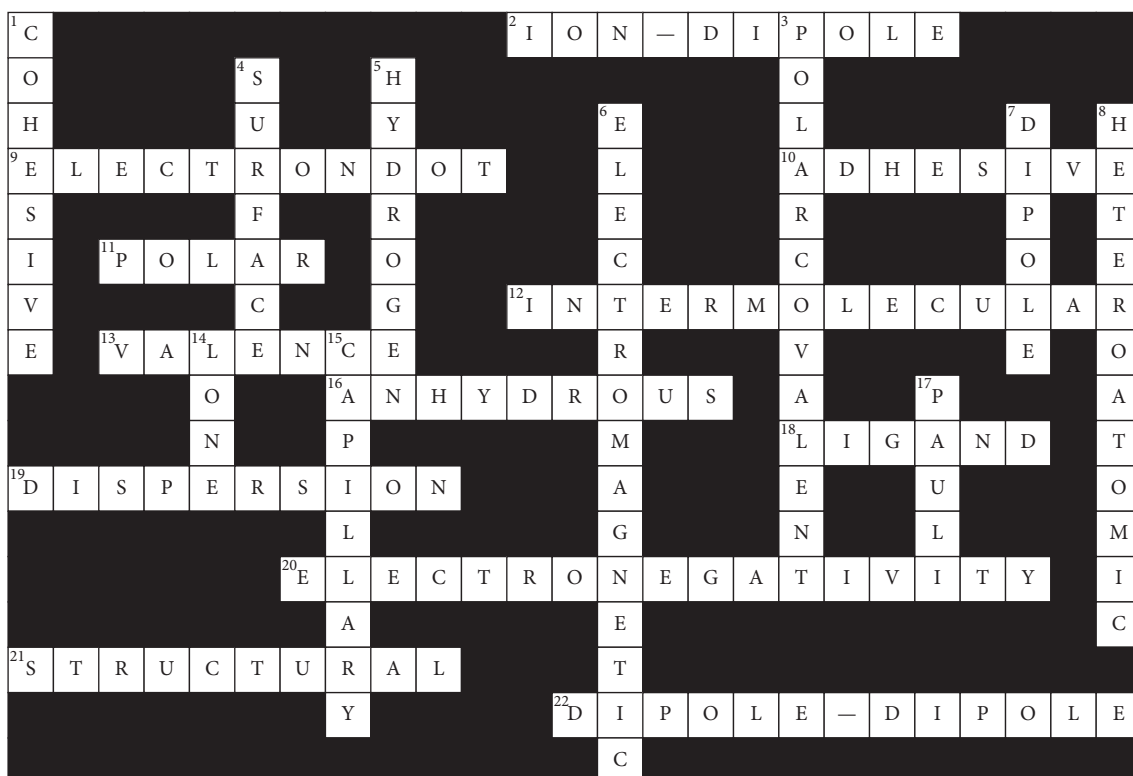
ACTIVITY SHEET ANSWERS

Chapter 6 Suggested answers

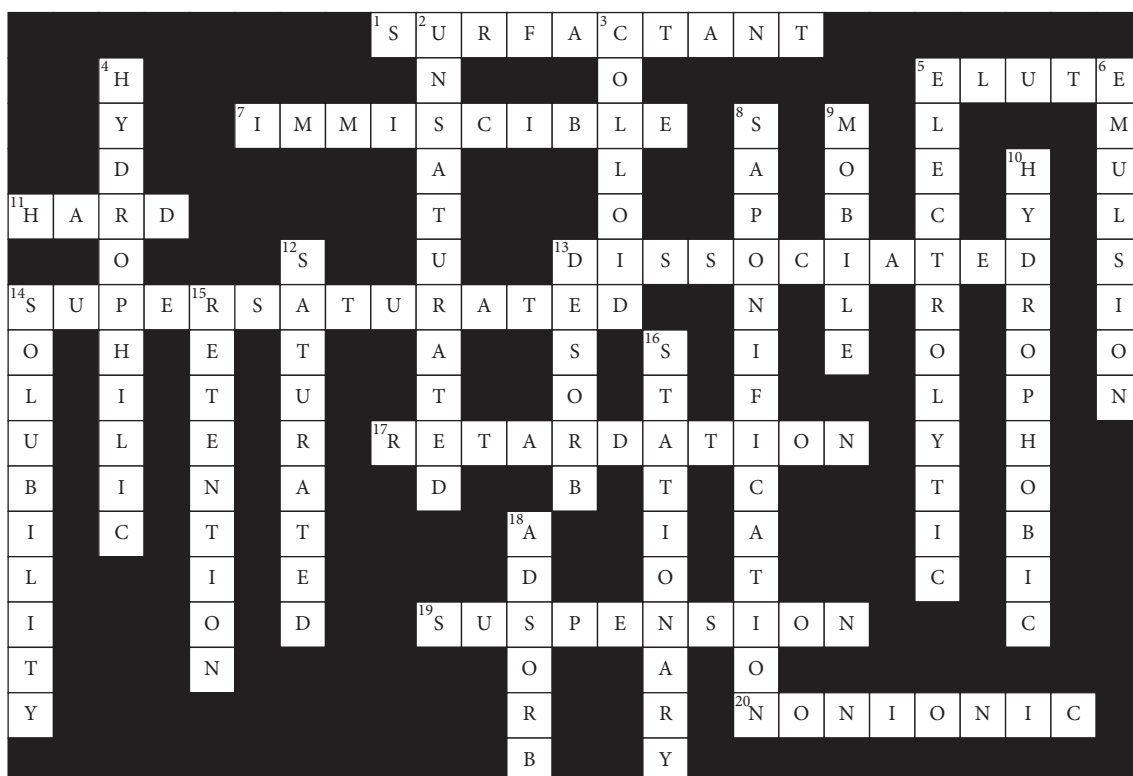
Prior knowledge

- 1 True
- 2 True
- 3 True
- 4 False: Electrons in a particular subshell all have *the same* amounts of energy.
- 5 True
- 6 False: Valence electrons are those found in the *outer* occupied shell of an atom.
- 7 True
- 8 False: Electrons in a particular subshell fill orbitals individually until they are forced to share an orbital, and only then if they have *opposite* spins.
- 9 True
- 10 True
- 11 False: Covalent bonding occurs when a pair of electrons is shared between two atoms because each gains a *more* energetically stable state.
- 12 True
- 13 True
- 14 False: A single bond occurs when a pair of atoms shares *one pair of electrons*, a double bond when they share *two pairs*, and a triple bond when they share *three pairs*.
- 15 True
- 16 True
- 17 False: Intermolecular forces are the forces of *attraction* that occur between the individual particles (molecules or ions) of a substance.
- 18 True
- 19 False: A *solute* is the substance that dissolves in a solvent to form a solution.
- 20 True

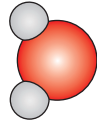
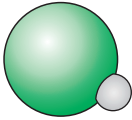
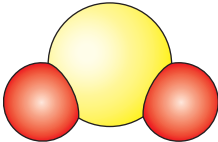
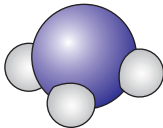
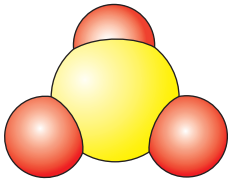
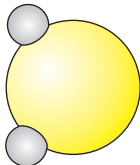
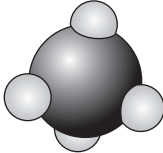
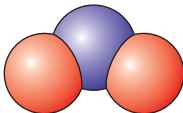
Activity sheet 6.1



Activity sheet 6.2



Activity sheet 6.3

Formula	Electron dot diagram	No. atoms bonded to central atom	No. of lone pairs on central atom	Space-filling model of molecule	Polarity of molecule
H ₂ O	$\begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ :\ddot{\text{O}}:\text{H} \\ :\ddot{\text{H}} \end{array}$	2	2		polar
HCl	$\begin{array}{c} \cdot\cdot \\ :\ddot{\text{Cl}}:\text{H} \\ \cdot\cdot \end{array}$	1	3		polar
SO ₂	$\begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ :\ddot{\text{S}}:\ddot{\text{O}} \\ :\ddot{\text{O}}:\cdot\cdot \end{array}$	2	1		polar
NH ₃	$\begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \text{H}:\ddot{\text{N}}:\text{H} \\ :\ddot{\text{H}} \end{array}$	3	1		polar
CO ₂	$\text{O}::\text{C}::\text{O}$	2	0		non-polar
SO ₃	$\begin{array}{c} \cdot\cdot \\ :\ddot{\text{O}}:\cdot\cdot \\ \cdot\cdot \\ :\ddot{\text{O}}:\ddot{\text{S}}:\ddot{\text{O}}:\cdot\cdot \\ \cdot\cdot \end{array}$	3	0		non-polar
H ₂ S	$\begin{array}{c} \cdot\cdot \\ :\ddot{\text{S}}:\text{H} \\ :\ddot{\text{H}} \end{array}$	2	2		polar
CH ₄	$\begin{array}{c} \text{H} \\ \cdot\cdot \\ \text{H}:\ddot{\text{C}}:\text{H} \\ \cdot\cdot \\ \text{H} \end{array}$	4	0		non-polar
NO ₂	$\begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ \text{N}:\ddot{\text{O}} \\ :\ddot{\text{O}}:\cdot\cdot \end{array}$	2	1		polar

Activity sheet 6.4

What does the equipment look like?	How does the technique work?	What does the output look like?	What is it used for?
Thin-layer chromatography	Solvent moves over stationary phase by capillary action. Components adsorb and desorb at different rates due to chemical structure and ability to form inter-molecular bonds to the stationary phase and mobile phase.		Organic mixtures such as plant pigments and drugs
Gas chromatography	Carrier gas is mobile phase. Sample is injected into an oven. The gas moves the vaporised particles through a fine long column. The smallest solutes that adsorb the least will elute first.	1 Lindane 2 Heptachlor 3 Aldrin 4 Dieldrin 5 DDT	Small, heat-stable organic compounds such as ethanol
Liquid chromatography	Similar to gas chromatography but mobile phase is a liquid. Unlike the gas in gas chromatography, the liquid plays an important role in adsorption/desorption process. A pump is needed to push the liquid through the denser packed column.		Larger volatile organic compounds

Chapter 6 Revision

Sections 6.1 to 6.5

- 1 a → L, b → G, c → J, d → B, e → K, f → S, g → M, h → V, i → T, j → A, k → O, l → D, m → N, n → C, o → R, p → I, q → Q, r → H, s → P, t → U, u → E, v → F

- 2 a no bonds: Ne
b 1 bond only: Cl, F, H
c up to two bonds: O, S
d up to 3 bonds: N, P
e up to 4 bonds: C, Si
f up to 5 bonds: none

3	Molecular formula	Molecular shape	Number of bonded atoms	Non-bonding pairs of electrons	Valence structure
	SO ₂	V-shaped	2	1	$\begin{array}{c} \ddot{\text{S}}=\ddot{\text{O}} \\ \\ \text{:O:} \end{array}$
	SO ₃	trigonal planar	3	0	$\begin{array}{c} \text{:}\ddot{\text{O}}-\text{S}-\ddot{\text{O}}\text{:} \\ \\ \text{:O:} \end{array}$
	CO ₂	linear	2	0	$\ddot{\text{O}}=\text{C}=\ddot{\text{O}}$
	H ₂ O	V-shaped	2	2	$\begin{array}{c} \text{:}\ddot{\text{O}}-\text{H} \\ \\ \text{H} \end{array}$
	NH ₃	trigonal pyramidal	3	1	$\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{H} \\ \\ \text{H} \end{array}$
	CH ₄	tetrahedral	4	0	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$
	NH ₄ ⁺	tetrahedral	4	0	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{N}-\text{H} \\ \\ \text{H} \end{array}$

4	Origin of force	Relative strength
Dipole-dipole	The slightly negative Cl end of polar HCl molecules attracts the slightly positive H ends of neighbouring HCl molecules and bonds form between them. Permanent dipoles thus attract each other.	Stronger than dispersion forces but weaker than hydrogen bonding
Hydrogen bonding	N, O and F are the most electronegative elements. When bonded to H, the bonding electrons are strongly attracted away from the H toward the electronegative atom. This results in the H orbital being largely empty so it can accommodate a share of a non-bonding electron pair from a neighbouring atom.	Strongest of the three forces
Dispersion	At any given instant, the orbiting electrons may be more concentrated in one region, and the molecule becomes an instantaneous dipole. It now induces a dipole on neighbouring molecules, which are attracted to it for an instant.	Weakest of the three forces

Sections 6.6 to 6.9

- 1 a $\rightarrow$ V, b $\rightarrow$ I, c $\rightarrow$ A, d $\rightarrow$ F, e $\rightarrow$ K, f $\rightarrow$ R, g $\rightarrow$ J, h $\rightarrow$ M, i $\rightarrow$ L, j $\rightarrow$ H, k $\rightarrow$ P, l $\rightarrow$ N, m $\rightarrow$ E, n $\rightarrow$ G, o $\rightarrow$ U, p $\rightarrow$ O, q $\rightarrow$ Q, r $\rightarrow$ B, s $\rightarrow$ T, t $\rightarrow$ W, u $\rightarrow$ C, v $\rightarrow$ D, w $\rightarrow$ Q
- 2 a The negatively-charged oxygen end of polar water molecules more strongly attract the positively charged magnesium ion than the negatively-charged chlorine ions in the lattice do.
b The negatively-charged chloride ion has been separated from the crystal lattice and is now surrounded by the positively-charged hydrogen ends of the polar water molecules.
c The positively-charged magnesium ion has been separated from the crystal lattice and is now surrounded by the negatively-charged oxygen ends of the polar water molecules.
- 3 a The only force that holds helium atoms together is very weak dispersion forces, so liquid helium boils at a very low temperature.
b The strong hydrogen bonding between molecules in the ice results in ice having a relatively high melting point.
c The attractive force between the highly-polar water molecules and the ions forming the salt is stronger than the forces between the ions themselves, so salt is highly soluble.
- 4 In general, at a higher temperature the salt will dissolve more rapidly because the water molecules move faster at a higher temperature and come into contact with the ions on the surface of the lattice more frequently, thus separating them more rapidly.
- 5 Heating a solution makes it easier for particles to move from the solid to the solution phase.
- 6 There are two phases: a stationary solid or liquid phase, and a mobile liquid or gas phase. The mobile phase flows through the stationary phase carrying the components of the mixture with it. As different components travel at different rates, they separate.

7

Technique	How does it work?	What is it used for?
Thin-layer chromatography	<i>Solvent moves over the stationary phase by capillary action. Components adsorb and desorb at different rates, thus separating.</i>	<i>Separating organic mixtures such as plant pigments and dyes</i>
Gas chromatography	<i>Sample is injected into an oven and a carrier gas moves vaporised particles through a long fine column, in which components separate.</i>	<i>Separating small heat-stable organic compounds</i>
High performance liquid chromatography	<i>The mobile phase is a liquid that carries the components through a stationary column in which the components separate.</i>	<i>Separating large complex organic compounds that are unstable when heated</i>